

Red Hat Lightspeed 1-latest

Assessing RHEL configuration issues by using the Red Hat Lightspeed advisor service with FedRAMP

Assess and monitor the configuration issues impacting your Red Hat Enterprise Linux systems

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Abstract

Use the Red Hat Lightspeed advisor service with FedRAMP® to assess and monitor configuration issues affecting the availability, stability, performance, and security of your RHEL systems.

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CHAPTER 1. IDENTIFY CONFIGURATION ISSUES ON YOUR RHEL SYSTEMS WITH ADVISOR

Use the advisor service to assess and monitor the health of your Red Hat Enterprise Linux infrastructure. The service identifies configuration issues affecting availability, stability, performance, and security across individual systems or your entire infrastructure.

1.1. HOW THE ADVISOR SERVICE WORKS

The advisor service runs daily to check RHEL systems against a database of recommendations on systems that have insights-client installed and registered. Each recommendation identifies risk conditions. You can view results in the Red Hat Hybrid Cloud Console to filter, analyze, and remediate issues with appropriate user permissions.

After the daily check runs, you can:

- See all of the recommendations for your entire RHEL infrastructure.
- Filter results by recommendations, systems, groups, or workloads. Supported filters include SAP workloads, Satellite host collections, and custom tags.
- View details about individual recommendations, including risks and resolutions tailored to your systems.
- Share results with other stakeholders. For more information, see Generating advisor service reports with FedRAMP.
- Create and manage remediation plans to fix issues in the Red Hat Lightspeed application. For more information, see Red Hat Lightspeed remediations guide with FedRAMP .

Additional resources

- Managing user permissions

CHAPTER 2. MANAGE USER PERMISSIONS FOR RED HAT LIGHTSPEED SERVICES

Manage user permissions to control access to Red Hat Lightspeed applications. Use the User Access feature to apply role-based access control (RBAC). Red Hat provides predefined groups and roles to help Organization Administrators manage user permissions.

2.1. USER ACCESS OVERVIEW

You can simplify access management in the Red Hat Hybrid Cloud Console by using role-based access control (RBAC). The User Access feature assigns specific permissions to roles instead of individual users, which are then inherited when roles are assigned to user groups.

You can also create custom groups and roles to provide more fine-tuned control over specific features of Red Hat Lightspeed to suit the needs of your organization.

If you are an Organization Administrator, you can use the User Access feature under **Identity & Access Management → User Access** in the Hybrid Cloud Console to:

- Control user permissions and organize roles.
- Create groups that include roles and their corresponding permissions.
- Assign users to these groups, allowing them to inherit the permissions associated with their group's roles.

All users on your account have access to most of the data in Red Hat Lightspeed.

Additional resources

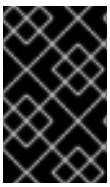
- User Access in the Hybrid Cloud Console
- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP

2.2. PREDEFINED GROUPS IN USER ACCESS

The two predefined groups available in User Access, **Default access** and **Default admin access** provide different baseline privilege levels for users in your organization. You can also create custom groups to align permissions with specific personas, job functions, or teams in your organization.

The Default access group

By default, the **Default access** group is assigned many granular predefined roles, so that group members have basic visibility. Because all users in your organization are members of the **Default access** group, they inherit all permissions assigned to that group. The **Default access** group is automatically updated by Red Hat.



IMPORTANT

If your Organization Administrator modifies the **Default access** group, the group is automatically renamed to **Custom default access**. Once converted, this group is no longer automatically updated by Red Hat.

The Default admin access group

The **Default admin access** group contains only users who have Organization Administrator permissions. This group is automatically maintained, and users and roles in this group cannot be changed.

The **Default admin access** group includes many (but not all) predefined roles that provide update and delete permissions. The roles in this group usually include **administrator** in their names.

TIP

To view the roles included in the **Default access** or **Default admin access** group, log in to the Hybrid Cloud Console, navigate to **Groups** in User Access, and select the group.

Additional resources

- User Access groups

2.3. PREDEFINED ROLES ASSIGNED TO GROUPS

Understand how predefined roles in Red Hat Hybrid Cloud Console bundle permissions across multiple Red Hat Lightspeed applications to align with common user personas. Use predefined roles to reduce administrative effort, or create custom roles for more fine-tuned control over specific features.

The predefined roles are a starting point to help you control and manage user permissions. You can then use these roles to create custom roles that are tailored to your specific use cases and organization. For example, you can use the predefined granular roles to create custom roles that provide more fine-tuned control over specific features of Red Hat Lightspeed.

TIP

Across the Red Hat Lightspeed product documentation, the **Prerequisites** section for each procedure lists which predefined roles provide the permissions needed to use the features in that procedure. For example, if a procedure requires permissions to create and view remediations, the **Prerequisites** section for that procedure lists the *Remediations user* or other valid role as a recommended predefined role to use for that procedure.

Additional resources

- Predefined User Access roles

2.4. CHECK YOUR PERMISSIONS

Verify your current permissions and the roles or groups assigned to you in the Red Hat Hybrid Cloud Console. Check your permissions to troubleshoot access issues or understand your level of access to Red Hat Lightspeed applications.



NOTE

Only users with the Organization Administrator role can view the permissions of other users in the User Access settings and manage user permissions to Red Hat Lightspeed services. For more information, see the *Configure user permissions* section.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ My User Access**.
2. If you try to access Red Hat Lightspeed features and see a message that you do not have permission to perform this action, contact your Organization Administrator or a user with the **User Access administrator** role to request the permissions required to access those features and complete the actions you want to perform.

Verification

All of the applications that you have permissions to access are listed on this page and are grouped by product, for example, RHEL, OpenShift Container Platform, and Ansible Automation Platform.

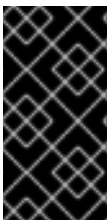
You can also filter your permissions by application, for example, by advisor, cost management, inventory, and remediations.

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

2.5. CONFIGURE USER PERMISSIONS

If you are an Organization Administrator, you can view and manage user permissions for all users in your organization. Control access to Red Hat Lightspeed and other Red Hat Hybrid Cloud Console services through the User Access interface.



IMPORTANT

If you are not an Organization Administrator, you are unable to complete this task. However, you can check your own permissions for different applications by navigating to **Settings → Identity & Access Management→ My User Access**. Contact your Organization Administrator to request more permissions.

Prerequisites

- You have logged in to the Red Hat Hybrid Cloud Console as an Organization Administrator, or you have the required administrator User Access role permissions.

Procedure

1. Click the **Settings** icon (⚙), then navigate to **Settings → Identity & Access Management→ User Access**.

Verification

From here, you can create and manage:

- **Settings → Identity & Access Management→ User Access → Roles** to determine permissions to Red Hat Lightspeed services and features

- **Settings → Identity & Access Management→ User Access → Groups** to include one or more roles to align with a specific persona, job function, or team in your organization
- **Settings → Identity & Access Management→ User Access → Users** and their assignment to groups to inherit permissions from the roles assigned to those groups

Additional resources

- User Access Configuration Guide for Role-based Access Control (RBAC) with FedRAMP
- Procedures for configuring User Access

2.6. USER ACCESS ROLES FOR PERMISSIONS TO THE ADVISOR SERVICE

Understand the predefined roles that control access to the advisor service of Red Hat Lightspeed. Use these role definitions to assign appropriate permissions to users based on their responsibilities.

The following table shows the standard and enhanced access permissions provided by the predefined roles in User Access for the advisor service:

Table 2.1. Permissions provided by the User Access roles

User Access role	Grants access to ...	Included in the Default access group
RHEL Advisor administrator	• Do any available operations on any RHEL advisor resource.	X
RHEL Advisor viewer	• Read RHEL advisor data.	

CHAPTER 3. VIEW AND MANAGE ADVISOR SERVICE RECOMMENDATIONS

View and manage advisor service recommendations that identify configuration issues affecting your RHEL systems. Recommendations help you fix issues that impact availability, stability, performance, or security.

3.1. ASSESS AND PRIORITIZE RECOMMENDATIONS

The advisor Recommendations page displays sortable and filterable information about recommendations, including type, severity, and affected systems. Use these fields to assess and prioritize remediations.

The Recommendations page shows the following information about advisor service recommendations:

- **Modified**. Shows the date or time frame when a recommendation was last modified (or published) to the advisor service database.
- **Category**. Shows the type of issue or what the issue affects—whether the issue has the potential to negatively affect the availability, stability, performance, or security of RHEL systems.
- **Impact**. Shows the impact (**Low**, **Medium**, **High** or **Critical**) on system operations if an incident related to the recommendation were to happen.
- **Incidents**. Shows the issue has been classified as an incident and is an issue that is already affecting your systems.
- **Likelihood**. Shows the likelihood (**Low**, **Medium**, **High** or **Critical**) that the issue will negatively affect your infrastructure.
- **Name**. Shows a brief description of the issue, including how it affects RHEL systems.
- **Reboot Required**. Shows whether a system reboot is required (**Required** or **Not Required**) as part of remediation steps.
- **Remediation type**. Shows if the fix or remediation requires manual steps or has an automated playbook.
- **Risk of change**. Shows the risk of change (**Very Low**, **Low**, **Moderate**, or **Important**) for systems if a remediation is executed.
- **Status**. Describes a recommendation's status. (**All**, **Enabled**, **Disabled**, or **Red Hat disabled**).
    - **All**. Shows results for all status options.
    - **Enabled** and **Disabled**. Show business-critical recommendations that are visible, or not visible. A disabled recommendation indicates that someone in your organization turned the visibility of a recommendation off.
    - **Red Hat disabled**. Shows recommendations that Red Hat makes available for you to enable when you are performing specific actions, such as upgrading Red Hat Enterprise Linux major versions.

- **Systems** or **Systems impacted**. Shows the number of systems (**1 or more** or **None**) on which a recommendation is detected. **1 or more** indicates systems with recommendations. **None** indicates systems without recommendations.

- **Total risk**. Describes total risk level (**Low**, **Moderate**, **Important**, or **Critical**) based on likelihood and impact. The risk assessment considers how likely the issue will affect your infrastructure and the severity of that impact.

The default advisor service recommendations view shows a smaller subset of information. The information is filtered to show recommendation information for one or more systems. These recommendations have a status of **Enabled**.

Additional resources

- Filtering advisor service recommendations

3.2. INFORMATION PROVIDED FOR AN ADVISOR RECOMMENDATION

When a recommendation exists for a system, the advisor service identifies whether and how the system has been affected and provides specific mitigation or resolution instructions. This information is visible when you view a recommendation and then select an affected system.

After selecting an affected system, view all recommendations available for the system along with the following information:

- **Detected issues** Specific information about the fault on that system

- **Steps to resolve** Steps to resolve the issue on *that* system

- **Related Knowledgebase articles** KB articles or solutions about the general issue

- **Additional info** Other support articles on the issue or solutions for resolution

- **Ansible** Ansible Playbook remediation availability

3.3. RECOMMENDATION IMPACT DATE

A recommendation *impacts* a system when the conditions it describes exist on that system. The advisor service shows you when a system first became impacted by a recommendation.

You can apply the **Systems impacted** primary filter to the list of recommendations, and select the **1 or more** secondary filter to show only recommendations impacting systems. By clicking on a recommendation from the filtered list, you can see when that recommendation first impacted each of the systems on the list.

You can view this information in the **First impacted** column on the following pages in the advisor service web console:

- On the **Recommendations** list for a single system, for each recommendation where applicable.

- In the details view of a single recommendation, **Affected systems** list, for each system impacted by that recommendation.

CHAPTER 4. REFINE ADVISOR SERVICE RECOMMENDATIONS

Refine advisor recommendations by viewing, filtering, sorting, and disabling them to focus on the issues most relevant to your systems.

4.1. VIEW ALL ADVISOR SERVICE RECOMMENDATIONS

When you first enter the advisor service Recommendations view, you see the default view and results of **Systems Impacted** and **Status** filters applied to the list of recommendations. By default, the **Systems Impacted** filter is set to **1 or more systems impacted**, and **Status** is set to **Enabled**.

To get a comprehensive view of *all* recommendations, including those not affecting your systems and those in the advisor database, close all the filters.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **RHEL Advisor viewer** role.

Procedure

1. Navigate to **Operations → Advisor → Recommendations**.
2. Click the **Close** icon  next to the **Systems Impacted** and **Status** filters. You can now browse through all of the potential recommendations for your systems.
3. Optional: Click **Reset filters** to return to the default recommendations view.

Additional resources

- Filter advisor service recommendations
- Disable an advisor service recommendation

4.2. FILTER ADVISOR SERVICE RECOMMENDATIONS

You can set filters and secondary filters to refine advisor service recommendations.

Use filters to focus on recommendations by severity, impact, category, or remediation type. Select from the following filters to refine your recommendations list:

- **Name.** In the secondary filter field, start typing the recommendation description or a keyword and select from the options presented.
- **Total risk.** In the secondary filter field, select from one or more: Critical, Important, Moderate, or Low.
- **Risk of change.** In the secondary filter field, select from High, Moderate, Low, or Very low.
- **Impact.** In the secondary filter field, select from Critical, Important, Moderate, or Low.
- **Likelihood.** In the secondary filter field, select from Critical, Important, Moderate, or Low.

- **Category.** In the secondary filter field, select from Availability, Performance, Stability, or Security.
- **Incidents.** In the secondary filter field, select to show recommendations with or without incidents having occurred.
- **Remediation.** In the secondary filter field, select Ansible Playbook or Manual for the remediation method.
- **Reboot required.** In the secondary filter field, select either Required or Not required.
- **Ansible support.** In the secondary filter field, select to show recommendations with or without Ansible Playbook support.
- **Status.** In the secondary filter field, select from All, Enabled, Disabled, Red Hat disabled.
- **Systems impacted.** In the secondary filter field, select either 1 or more or None.

Procedure

1. Navigate to **Operations → Advisor → Recommendations**.
2. Click the filter icon and select a filter category from the drop-down list.
3. Click the drop-down arrow in the secondary filter menu and select the required options to activate a secondary filter.



NOTE

For some filter options, for example **Name**, type the name or description of a recommendation to refine the list further.

Additional resources

- View all advisor service recommendations
- Filter systems by tags and workloads

4.3. PARAMETERS FOR SORTING ADVISOR RECOMMENDATIONS

You can sort the advisor recommendations table by name, date modified, total risk level, number of affected systems, or whether Ansible Playbook remediation is available. Sorting by total risk helps you prioritize critical issues, while sorting by number of systems shows which recommendations affect the most infrastructure.

- **Name.** Alphabetize by A to Z or Z to A.
- **Modified.** Order by number of days since the recommendation was last modified or published, from newest or oldest.
- **Total risk.** View in order of criticality.
- **Systems.** View by the number of your systems that are impacted.
- **Remediation.** Sort by recommendations that have or do not have Ansible Playbook support.

4.4. DISABLE AN ADVISOR SERVICE RECOMMENDATION

You can disable specific recommendations that affect your systems so that they no longer appear in your results.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **RHEL Advisor administrator** role.

Procedure

1. In the Hybrid Cloud Console, navigate to the Operations > Advisor > Recommendations page.
2. Locate the recommendation to disable.
3. Click **More-options** () at the right end of the row, and then click **Disable recommendation**. A confirmation dialog is displayed.
4. Under **Justification note**, enter a comment to explain why you are disabling the recommendation, and then click **Save**.

Results

The recommendation is disabled and is no longer displayed in your reports and dashboards.

Additional resources

- View and enable a previously disabled recommendation
- View all advisor service recommendations

4.4.1. View and enable a previously disabled recommendation

When a recommendation is disabled, you no longer see the recommendation in your advisor results. To reverse this action, use the **Status** filter.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **RHEL Advisor administrator** role.

Procedure

1. Navigate to **Operations → Advisor → Recommendations**.
2. Click the Filter drop-down and select **Status**.
3. In the secondary filter drop-down list, select **Disabled**.
4. Locate the recommendation to enable.

5. Navigate to the right side of the row, and click the **More-options** icon  followed by **Enable recommendation**.

Additional resources

- Disable an advisor service recommendation
- View all advisor service recommendations

CHAPTER 5. REFINE YOUR VIEW OF SYSTEMS IN THE ADVISOR SERVICE

Filter and sort systems in the advisor service by name, tags, or other attributes to focus on specific systems that are reporting configuration data.

5.1. INDIVIDUAL SYSTEM RECOMMENDATIONS

Use the search filter to locate specific systems by hostname in the advisor service recommendations table.

5.2. OPTIONS FOR SORTING RHEL SYSTEMS

The systems table supports sorting by name, number of recommendations, and last-seen status to help you organize and prioritize systems that require attention.

- **Name.** Alphabetize by A to Z or Z to A.
- **Number of recommendations.** Order by the number of recommendations impacting each system.
- **Last seen.** Order by the number of minutes, hours, or days since an archive was last uploaded from the system to the advisor service.

5.3. FILTER SYSTEMS BY TAGS AND WORKLOADS

To quickly locate and view the systems you want to focus on, filter results in the advisor service UI by custom group tags, SAP workloads, and Satellite groups.

In the advisor service, use the **Filter results** box to access tag, workload, and group filters. The **Filter** drop-down menu shows all of the tags associated with the account. Click one or more parameters to apply filters.

Procedure

1. Navigate to **Operations → Advisor → Systems**. The **Filter results** box is in most views in the Red Hat Lightspeed application and these procedures work anywhere you access **Filter results**.
2. Click the arrow on the **Filter results** box and scroll to see the tags available for systems on this account.
3. Select one or more tags to filter by SAP workloads, Satellite host group, or a custom group. Applied tags are visible next to the **Filter results** box.
4. View the filtered results throughout the advisor service.
5. To remove the tag, click **Clear filters**.

Additional resources

- Organize systems in Red Hat Lightspeed by using filtering and tags

CHAPTER 6. ORGANIZE SYSTEMS IN RED HAT LIGHTSPEED BY USING FILTERING AND TAGS

Filter and organize your systems in Red Hat Lightspeed by using SAP workload tags, Satellite host group tags, and custom tags. Tagging enables you to focus on specific system groups across multiple services, which simplifies management for large-scale deployments.

6.1. FILTERS AND TAGS FOR SAP WORKLOADS

Red Hat Lightspeed automatically tags systems by SAP workload type and SAP ID (SID), which you can use to filter and organize SAP systems.

Use these automatic tags to filter and organize your SAP systems throughout the Red Hat Lightspeed application. No customization or manual tagging is required. To filter to view your SAP workloads, use **Workloads > SAP**.

6.2. FILTERS AND TAGS FOR SATELLITE HOST GROUPS

Satellite host groups are configured in Satellite and are automatically recognized by Red Hat Lightspeed.

6.3. CUSTOM SYSTEM TAGGING

Apply custom tags to your registered systems to add contextual markers and organize your Red Hat Lightspeed inventory. With custom tags, you can filter and focus on related groups of systems, which simplifies management for large-scale deployments.

You can also apply predefined tags to your systems. The advisor service uses predefined tags to generate targeted recommendations for systems that require specific attention, such as those with high security needs.

6.3.1. Filter structure

Filters use a **namespace=value** or **key=value** paired structure.

- **Namespace.** The namespace is the name of the ingestion point, *insights-client*. This value cannot be changed. The **tags.yaml** file is abstracted from the namespace, which is injected by the client before upload.
- **Key.** You can create the key or use a predefined key from the system. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.
- **Value.** You can define your own descriptive string value. You can use a mix of capitalization, letters, numbers, symbols, and whitespace.

6.3.2. Create a custom group and the tags.yaml file

To create and add tags to **/etc/insights-client/tags.yaml**, use **insights-client** with the **--group=<name-you-choose>** option. Running this command on your systems that are registered to Red Hat Lightspeed completes the following actions:

- Creates the **etc/insights-client/tags.yaml** file

- Adds the **group=** key and **<name-you-choose>** value to **tags.yaml**
- Uploads a fresh archive from the system to the Red Hat Lightspeed service, making the new tag immediately visible along with your latest results

Prerequisites

- Root-level access to your system.

Procedure

1. Run the following command as root, adding your custom group name in place of **<name-you-choose>**:

```
[root@server ~]# insights-client --group=<name-you-choose>
```

2. Optional: To add additional tags, edit the **/etc/insights-client/tags.yaml** file.
3. Navigate to **Inventory → Systems** and log in if necessary.
4. Click the **Filter by tags** drop-down menu. You can also use the search box to enter all or part of the tag's name to automatically show systems with that text in the tags.
5. Scroll up or down the list to locate the tag.
6. Click the tag to filter by it.
7. Verify that your system is among the results on the advisor systems list.
   a. Navigate to **Inventory → Systems** and log in if necessary.
   b. Activate the **Name** filter and begin typing the system name until you see your system, then select it.
   c. The tag symbol is a darker color, and the number beside it shows the correct number of tags applied.

6.3.3. Add and modify custom system tags by editing tags.yaml

Edit the **tags.yaml** file to add or modify custom system tags after creating a **group** tag. Using custom system tags can help you streamline the organization and filtering of systems when you use Red Hat Lightspeed inventory and other services.

Prerequisites

- Root-level access to your system.

Procedure

1. Open the tag configuration file, **tags.yaml**, in an editor.

```
[root@server ~]# vim /etc/insights-client/tags.yaml
```

2. Edit the file contents or add **key=value** pairs as needed. Use a mix of capitalization, letters, numbers, symbols, and whitespace. Example **tags.yaml** with multiple tags:

```
group: _group-name-value_
location: _location-name-value_
description:
- RHEL8
- SAP
key 4: value
```

3. Save your changes and close the editor.

4. Initiate an upload to Red Hat Lightspeed.

```
[root@server ~]# insights-client
```

5. Navigate to **Inventory → Systems** and log in if necessary.

6. In the **Filter Results** box, click the down arrow and select one of the filters or enter the name of the filter and select it.



NOTE

You can search by the tag key or by its value.

7. Find your system among the results.

8. Verify that the filter icon is highlighted and shows a number representing the number of filters applied to the system.

6.4. PREDEFINED TAGS

In addition to custom tags, Red Hat Lightspeed provides predefined tags that the advisor service can use to create targeted recommendations. These predefined tags are useful for systems that require more security or need different networking performance levels.

To get the extended security hardening and enhanced detection and remediation capabilities offered by predefined tags, opt in and configure the tags. After configuration, the advisor service provides recommendations based on tailored severity levels and the preferred network performance that applies to your systems.

Use the **/etc/insights-client/tags.yaml** file to tag systems with predefined tags, similar to how you tag systems in the inventory service. The predefined tags are configured using the same **key=value** structure used to create custom tags. Details about the Red Hat-predefined tags are in the following table:

Table 6.1. List of supported predefined tags

Key	Value	Note

Key	Value	Note
security	**normal** (default) / **strict**	With the **normal** (default) value, the advisor service compares the system's risk profile to a baseline that comes from the default configuration of the most recent version of RHEL and from often-used usage patterns. This option keeps recommendations focused, actionable, and low in numbers. The **strict** value is best for systems that are security-sensitive. This option uses a stricter baseline and can show recommendations even on the most up-to-date RHEL installations.
network_performance	**null** (default) / **latency** / **throughput**	The network performance values, either latency or throughput according to your business requirement, affect the severity of an advisor service recommendation to a system.



NOTE

The names of predefined tag keys are reserved for specific use. If you already use the key **security**, with a value that differs from one of the predefined values, you do not see a change in your recommendations. You only see a change in recommendations if your existing **key=value** is the same as one of the predefined keys. For example, if you have a **key=value** of **security: high**, your recommendations do not change because of the predefined tags that Red Hat has already defined. If you currently have a **key=value** pair of **security: strict**, you see a change in the recommendations for your systems.

Additional resources

- Using system tags to enable extended security hardening recommendations
- Leverage tags to make Red Hat Lightspeed Advisor recommendations understand your environment better
- Custom system tagging

6.4.1. Configure predefined tags

Configure predefined tags to adjust recommendations and gain extended security hardening, enhanced detection, and remediation capabilities within the Red Hat Lightspeed advisor service.

Prerequisites

- You have root-level access to your system.
- You have insights-client installed.
- You have systems registered within the insights-client.
- You have created the **tags.yaml** file. For information about creating the **tags.yaml** file, see Configure custom group tags.

Procedure

1. Use the command line and your preferred editor to open **/etc/insights-client/tags.yaml**. (The following example uses Vim.)

```
[root@server ~]# vi /etc/insights-client/tags.yaml
```

2. Edit the **/etc/insights-client/tags.yaml** file to add the predefined **key=value** pair for the tags. This example shows how to add **security: strict** and **network_performance: latency** tags.

```
# cat /etc/insights-client/tags.yaml
group: redhat
location: Brisbane/Australia
description:
- RHEL8
- SAP
security: strict
network_performance: latency
```

3. Save your changes.
4. Close the editor.
5. Optional: Run the **insights-client** command to generate an upload to Red Hat Lightspeed, or wait until the next scheduled Red Hat Lightspeed upload.

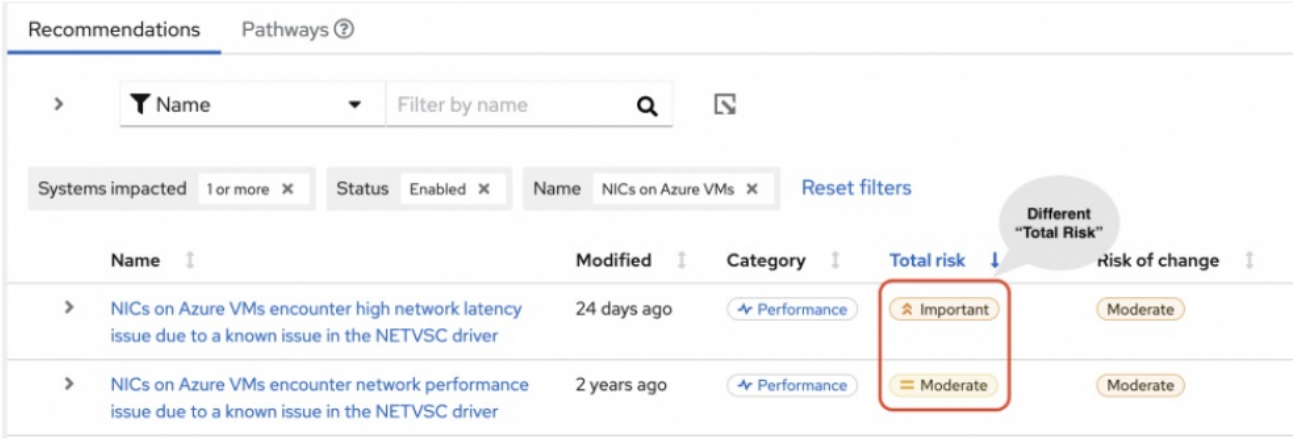
```
[root@server ~]# insights-client
```

Verification

After generating an upload to Red Hat Lightspeed (or waiting for the next scheduled Red Hat Lightspeed upload), you can discover whether the tags are in the production environment by accessing Red Hat Lightspeed > RHEL > Inventory > Systems . Find your system and look for the newly created tags. A table displays:

- Name
- Value
- Tag Source (for example, insights-client).

The following image of the advisor service shows a system with the **network_performance: latency** tag configured.



The system shows a recommendation with a higher Total Risk level of Important. The system without the **network_performance: latency** tag has a Total Risk of Moderate. You can make decisions about prioritizing the system with higher Total Risk.

6.5. ADDITIONAL RESOURCES

- Refining your view of systems in inventory
- Enable SQL Server recommendations in Advisor
- Group and tag filters in the compliance service

CHAPTER 7. ENABLE SQL SERVER RECOMMENDATIONS IN ADVISOR

The advisor service provides recommendations for Microsoft SQL Server running on Red Hat Enterprise Linux to identify configuration issues that can affect performance, security, and stability. The SQL Assessment API evaluates database configuration for best practices by using rules from Microsoft. The insights-client uploads assessment results as JSON files and presents them in the Red Hat Lightspeed UI.

7.1. MICROSOFT SQL SERVER RECOMMENDATIONS IN ADVISOR

The Red Hat Lightspeed advisor service helps administrators identify configuration issues on Microsoft SQL Server systems running on RHEL 8.5 or later. The service integrates the Microsoft SQL Assessment API with operating system analysis to detect issues affecting performance, security, and stability.

The SQL Assessment API evaluates the database configuration of Microsoft SQL Server for best practices. The API includes a rule set with best practice rules from the Microsoft SQL Server Team. New versions enhance this rule set. The API provides a customizable and extensible solution that lets you tune the default rules and create your own.

Microsoft provides a PowerShell script (supported by PowerShell for Linux) that calls the Assessment API and stores results as JSON. The insights-client uploads this file from each system and transforms it into an easy-to-understand visual format in the Red Hat Lightspeed UI. Instead of parsing raw JSON data, you see clear recommendations with context about the systems that need attention and what to fix.

Additional resources

- SQL Server database best practices now available through Red Hat Lightspeed

7.2. CONFIGURE SQL ASSESSMENT API FOR ADVISOR RECOMMENDATIONS

Configure the Microsoft SQL Assessment API to send database assessment information to Red Hat Lightspeed. You create authentication credentials, install PowerShell and the **SqlServer** module, download the assessment script, and run your first assessment.

Procedure

1. In the database you want to assess, create a login for SQL Server assessments using SQL Authentication. The following Transact-SQL creates a login. Replace **<PASSWORD>** with a strong password:

```
USE [master]
GO
CREATE LOGIN [assessmentLogin] with PASSWORD= N'<PASSWORD>'
GO
ALTER SERVER ROLE [sysadmin] ADD MEMBER [assessmentLogin]
GO
```

2. Store and secure the credentials for login on the system. Replace *<PASSWORD>* with the password you used in step 1.

```
# echo "assessmentLogin" > /var/opt/mssql/secrets/assessment
# echo "<PASSWORD>" >> /var/opt/mssql/secrets/assessment
# chmod 0600 /var/opt/mssql/secrets/assessment
# chown mssql:mssql /var/opt/mssql/secrets/assessment
```

3. Download PowerShell from the Microsoft package repository. This is the same repository you configured when you installed the **mssql-tools** and **mssqlodbc17** packages as part of SQL Server installation.

```
# yum -y  install powershell
```

4. Install the **SqlServer** module for PowerShell. The module includes the assessment API.

```
# su mssql -c "/usr/bin/pwsh -Command Install-Module SqlServer"
```

5. Download the **runassessment** script from the Microsoft examples GitHub repository. Ensure the **mssql** user owns the file and can run it.

```
# /bin/curl -LJ0 -o /opt/mssql/bin/runassessment.ps1
https://raw.githubusercontent.com/microsoft/sql-server-samples/master/samples/manage/sql-assessment-api/RHEL/runassessment.ps1
# chown mssql:mssql /opt/mssql/bin/runassessment.ps1
# chmod 0700 /opt/mssql/bin/runassessment.ps1
```

6. Create the directory that stores the log file used by Red Hat Lightspeed. Ensure the **mssql** user owns the directory and can access it.

```
# mkdir /var/opt/mssql/log/assessments/
# chown mssql:mssql /var/opt/mssql/log/assessments/
# chmod 0700 /var/opt/mssql/log/assessments/
```

7. Create your first assessment as the **mssql** user. This allows later assessments to run automatically and securely through **cron** or **systemd**.

```
# su mssql -c "pwsh -File /opt/mssql/bin/runassessment.ps1"
```

8. Red Hat Lightspeed automatically includes the assessment the next time it runs, or you can start the insights-client by running this command:

```
# insights-client
```

9. Optional: Set up the SQL Assessment API call to run on a timer.
Because SQL Server assessments can take 10 minutes or more to complete, consider whether automatic assessments are appropriate for your environment. If you want to run assessments automatically, use the systemd service and timer files provided by the Red Hat SQL Server community.

   a. Download **mssql-runassessment.service** and **mssql-runassessment.timer** from the Red Hat SQL Server Community of Practice GitHub site to your current directory.

   b. On the command line, copy the files to /**etc**/**systemd**/**system**/:

```
# cp mssql-runassessment.service /etc/systemd/system/
# cp mssql-runassessment.timer /etc/systemd/system/
```

c. Set the file permissions:

```
# chmod 644 /etc/systemd/system/mssql-runassessment.service
/etc/systemd/system/mssql-runassessment.timer
```

d. Enable the timer:

```
# systemctl enable --now mssql-runassessment.timer
```

CHAPTER 8. USE PATHWAYS TO RESOLVE MULTIPLE ADVISOR SERVICE RECOMMENDATIONS

A *pathway* is a group of advisor-service recommendations that share a common resolution. Pathways present clear paths to follow to maintain your systems more efficiently.

8.1. HOW PATHWAYS WORK

By following the remediation steps in a pathway, you can view and address multiple recommendations in one configuration change. Use pathways to see all of the systems that the remediation affects without having to investigate every action within every recommendation.

The specific recommendations that appear in a pathway depend on the issues that affect your infrastructure at a given time. Red Hat Lightspeed dynamically calculates remediation levels based on the most significant recommendations in your environment. The calculation includes the following factors:

- Number of systems impacted by individual recommendations
- Total risk of individual recommendations
- Whether there is an incident detected

Issues grouped into pathways must share the same common resolution type, such as a package update, configuration update, product upgrade, and so on. In addition, the issues must share a common resolution target. Each system must require the same package update or configuration file change.

Red Hat Lightspeed names each pathway after its core remediation, so you can immediately see and understand the configuration change.

Additional resources

- Refining advisor service recommendations
- Red Hat Lightspeed remediations guide with FedRAMP

8.2. VIEW AND ASSESS ADVISOR PATHWAYS

View the Recommended Pathways panel to identify the most critical recommendations affecting your systems. Pathways shows you related recommendations grouped together by priority to help you focus remediation efforts where they matter most.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **RHEL Advisor viewer** role.

Procedure

1. Navigate to **Operations → Advisor → Recommendations**. The **Recommended Pathways** panel appears at the top of the page, and includes up to three of the most significant pathways for your systems.

2. To view details for a pathway, click **View Pathway**, or click **Pathways** in the list of recommendations and then select the pathway from the list. The information for the pathway is displayed.

3. Click **Systems** to view the list of systems that the issue affects.

Next steps

- After you view the pathways, decide which recommendations you want to add to a remediation plan. Each advisor service issue requires 20 action points for each affected system. Depending on the number of systems that the pathway affects, your plan could quickly exceed the 1000-point hosted execution limit. This means that you would not be able to implement the remediation plan directly on Red Hat Lightspeed, but you can download the plan and run it by using Red Hat Ansible Automation Platform (AAP).

Additional resources

- How pathways work
- Remediate pathways

8.3. REMEDIATE ADVISOR PATHWAYS

Use pathways to perform remediations on multiple systems at a time. You can select the affected systems from a list, and then remediate using Ansible playbooks or manual remediations.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **RHEL Advisor administrator** role.



NOTE

If the remediation shows **System reboot is required**, Ansible Automation Platform automatically reboots the affected system (or systems) after the remediation is complete.

Procedure

1. Navigate to **Operations → Advisor → Recommendations**.

2. Locate the pathway you want to remediate and click **View Pathway**.

3. Select a recommendation to remediate, or click **Systems** to see a list of all of the affected systems.

4. To view more information about each recommendation, expand the table row for the recommendation, or click the recommendation title and navigate to the **Recommendation Details** page.

5. Select the systems you want to remediate, and then click **Plan remediation**.

    a. If the remediation requires an Ansible Automation Platform playbook, follow the steps in the **Remediate with Ansible** dialog box to create or select a playbook and select the systems to remediate.

b. If the remediation requires manual remediation, follow the **Steps to resolve** procedure to perform the remediation on each of the selected systems.

Additional resources

- Red Hat Lightspeed remediations guide with FedRAMP
- Integrate with Red Hat Lightspeed (formerly Insights)
- Configure automation controller

8.4. CONFIGURE NOTIFICATIONS FOR ADVISOR RECOMMENDATIONS

Configure the notifications service to receive alerts when new advisor recommendations become available for your systems. You can receive notifications for individual recommendations in a pathway, but not for the pathway itself.

Prerequisites

- You are logged in to the Red Hat Hybrid Cloud Console as a user who is a member of a User Access group with the **RHEL Advisor administrator** role.

Procedure

1. Navigate to **Settings → Notifications → Red Hat Enterprise Linux**. The available behavior groups display.
2. Select an existing behavior group or create a new one.
3. Click the **More-options** ( ) icon for the behavior group, and select **Edit**. The **Edit behavior group** wizard is displayed.
4. Review the behavior group name and click **Next**. The **Actions and recipients** page is displayed.
5. Review the actions and recipients and click **Next**. The **Associate event types** page is displayed.
6. Select **New recommendation** to add it to the behavior group. Click **Next**.
7. Review the updated settings for the behavior group and click **Finish**.

Additional resources

- Configuring notifications on the Red Hat Hybrid Cloud Console with FedRAMP

CHAPTER 9. DELETE A SYSTEM FROM RED HAT LIGHTSPEED AND STOP DATA UPLOADS

Delete systems from inventory to stop them from uploading data to Red Hat Lightspeed. Deleting the system unregisters insights-client.

9.1. DELETE A SYSTEM FROM THE RED HAT SATELLITE UI

When you use Satellite to delete a system, you unregister the system from Satellite and remove the system from Red Hat Lightspeed system inventory.

Prerequisites

- You have root permissions for the system you want to delete.
- You are logged in to the Satellite web UI.

Procedure

1. Navigate to **Red Hat Lightspeed → Inventory**.
2. Select the system profile for the system that you want to remove.
3. Click **Actions** > **Unregister**.

Additional resources

- Red Hat Satellite product documentation

CHAPTER 10. REFERENCE MATERIALS

Access additional Red Hat Lightspeed documentation and support resources for remediations, reports, and product information.

For more information about Red Hat Lightspeed, see the following resources:

- Red Hat Lightspeed remediations guide with FedRAMP
- Generating advisor service reports with FedRAMP
- Red Hat Lightspeed Documentation
- Red Hat Lightspeed Product Support page